Agentic Multimodal RAG: Roles, Decisions, and Evaluation

Tianyu Yang[1,*], **Wenliang Guo**[2,*], **Xiujin Liu**[3], **Shijian Deng**[4], **Yilun Zhao**[5], **Yapeng Tian**[4], **Xiangliang Zhang**[1]

[1]University of Notre Dame
[2]Michigan State University
[3]University of Michigan, Ann Arbor
[4]The University of Texas at Dallas
[5]Yale University
[*]Equal contribution.

Multimodal retrieval-augmented generation (MM-RAG) enhances vision–language models by grounding generation in external multimodal knowledge. While prior work primarily focuses on *static* MM-RAG pipelines with fixed retrieval strategies, such approaches struggle with complex queries that require adaptive reasoning, and iterative evidence gathering. Recent advances have introduced **agentic MM-RAG**, where agents perform goal-driven, adaptive, and iterative decision-making throughout retrieval. Our paper provides an agent-centric synthesis of agentic MM-RAG, organizing the literature around four core agent functions: retrieval formulation, routing and retrieval, evidence verification in generation, and stopping under budget constraints. We further review function-aligned evaluation methodologies and discuss emerging applications, open challenges, and future research directions.

Project Page: https://github.com/Tianyu-yang-anna/Awesome-Agentic-Multimodal-RAG

1 Introduction

Large Vision Language Models (LVLMs) have emerged as a foundational paradigm for multimodal understanding and generation [5, 15, 35]. Yet, due to the limited coverage and timeliness of training corpora, purely parametric knowledge in LVLMs often falls short of meeting up-to-date, long-tail, or domain specific factual needs, and can lead to hallucinated or non traceable outputs [33]. Multimodal Retrieval Augmented Generation (MM-RAG) addresses this limitation by retrieving multimodal evidence from external knowledge sources and conditioning generation on the retrieved context, thereby improving factual grounding, updateability, and reliability [7, 25]. Despite these advances, most MM-RAG systems still follow a static, single step retrieve then generate pipeline. In challenging settings—such as ambiguous queries [37], multi-turn or multi-hop information needs, or sparse and dispersed evidence across heterogeneous sources [55]—the retrieval stage lacks explicit reasoning and interaction control, leading to incomplete coverage or coarse localization. These issues propagate to generation, ultimately limiting the credibility and robustness of the final response.

To address these limitations, recent work has begun to incorporate agentic frameworks [72] into MM-RAG. Agentic MM-RAG [24, 42, 75] reframes retrieval augmented generation as an *iterative* process that supports multi-step reasoning, dynamic tool use, and repeated evidence acquisition, as shown in Fig. 2. This paradigm improves adaptability and robustness under ambiguous and information intensive settings, and has shown promise in scientific discovery [52], medical assistance [65], and educational applications [29]. Meanwhile, agentization introduces new open challenges, including the reliability of long horizon reasoning and interaction, increased complexity and controllability concerns, and the safety, governance, and auditability of tool use and data access.

Motivated by the rapid progress of agentic MM-RAG, our framework extends the scope of existing *static* MM-RAG framworks that have not covered the distinctive roles agents play in interactive settings. Agents fundamentally reshape MM-RAG by enabling goal-driven, adaptive, and iterative decision-making throughout retrieval and generation. Accordingly, we organize this framework around the functional roles of agents, **1)** examining how they identify retrieval targets, **2)** decide what to retrieve and from where, **3)** integrate multimodal evidence into generation, and **4)** terminate iterative retrieval under budget constraints.

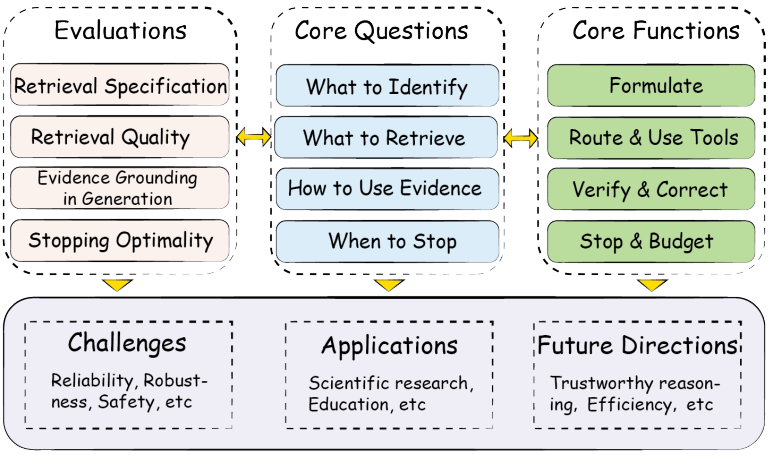


Figure 1 The roadmap of our framework.

As illustrated in Fig. 1, addressing these core questions requires agents to perform four core functions: **2) Formulate**, in which agents operationalize a query into explicit retrieval targets (e.g., sub-questions, modalities, and granularity); **2) Route and Use Tools**, in which agents select appropriate knowledge sources and retrieval mechanisms, execute retrieval actions, and locate relevant multimodal evidence; **3) Verify and Correct**. This involves assessing consistency between retrieved evidence and generated outputs, ensuring grounding, traceable citations, and alignment between claims and sources; and **4) Stop and Budget**, deciding whether to continue or terminate iterative retrieval-generation cycles under cost constraints.

This agent-centric organization contrasts with prior framework that structure *static* MM-RAG by task formulations or decomposed pipeline components [1, 41]. Our framework provides a unifying lens for synthesizing recent advances in agentic MM-RAG, covering agent decision-making throughout retrieval and generation, associated evaluation considerations, in dimensions of *retrieval specification, retrieval content quality, evidence grounding in generation*, and *stopping optimality under budget constraints*. This perspective further facilitates a structured discussion of open challenges, emerging applications, and future research directions.

The remainder of this framework is organized as follows. Section 2 introduces the problem formulation of agentic MM-RAG and contrasts it with the static MM-RAG setting. Section 3 then analyzes the roles of agents across the four core functions, with Table 1 summarizing and comparing representative works. Section 4 reviews evaluation methodologies for agentic MM-RAG across different dimensions, and Table 2 presents commonly used benchmarks and key metrics. Finally, Section 5 discusses emerging applications, open challenges, and future research directions.

2 Problem Setup

Static MM-RAG. A static MM-RAG system follows a fixed retrieve-then-generate pipeline. Given a user query, it encodes the query and candidate items from an external corpus into a shared embedding space, retrieves evidence via similarity matching, and generates a response conditioned on the query and retrieved evidence. Multimodal signals may enter the pipeline at different points, including the query, accessible knowledge sources, retrieved evidence, and output. Query and source modalities determine what the system can observe; evidence modality and granularity determine what is retrieved and grounded; and the output ranges from text-only to interleaved multimodal responses with explicit anchors.

Static MM-RAG aligns with a *fast-thinking* regime [17]: retrieval is executed without explicit deliberation and generation proceeds immediately once evidence is returned. This regime works when the information need is well specified and relevant evidence is accessible through a stable routine. However, in realistic multimodal settings, queries are often underspecified, key entities or attributes may be missing from partial visual context,

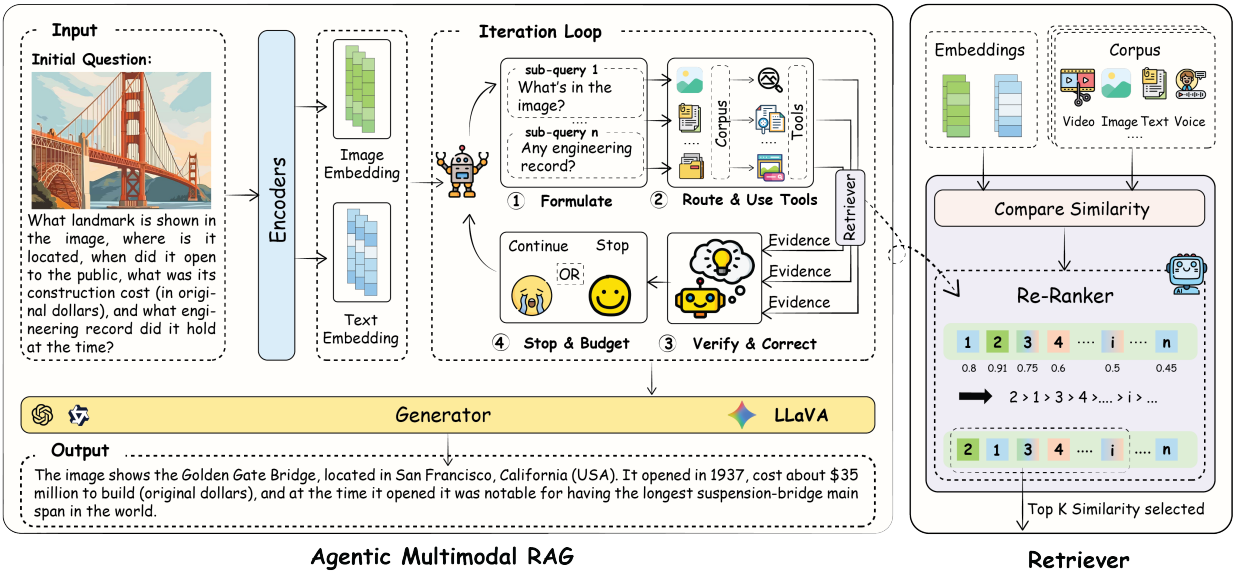


Figure 2 Overview of the agentic MM-RAG pipeline.

and evidence can be scattered across heterogeneous sources and tools. The appropriate retrieval modality and granularity may also be unknown upfront, so a single retrieval step can miss essential support or inject distracting context [53]. More fundamentally, the static abstraction offers limited control over query revision, source and tool switching, intermediate verification, and budgeted stopping. These limitations motivate a *slow-thinking* view in which retrieval becomes a deliberate, policy-controlled decision process, i.e., *Agentic MM-RAG*.

Agentic MM-RAG. Agentic MM-RAG is characterized by an iterative interaction between a policy-driven agent and an external environment that provides knowledge and tool capabilities. As illustrated in Fig. 2, the agent operates over embedded query representations to retrieve multimodal evidence, which is subsequently incorporated by the generator to produce grounded outputs. Formally, let $\mathcal{M}$ denote the set of supported modalities, and let a multimodal query be $q \in \mathcal{Q}(\mathcal{M})$. The environment $\mathcal{E}$ exposes heterogeneous knowledge sources and tool interfaces such as web and document collections, vector stores, knowledge graphs, tables, and modality-specific analyzers. At step t, the agent maintains a state $s_t \in \mathcal{S}$ that summarizes the evidence retrieved so far, intermediate hypotheses, the interaction trace, and remaining budgets. The agent selects an action $a_t \in \mathcal{A}$ under a policy π_θ, $a_t \sim \pi_\theta(\cdot \mid s_t)$. The environment executes a_t and returns an observation $o_t = \mathcal{E}(a_t)$, and the agent updates its state as $s_{t+1} = U(s_t, a_t, o_t)$. We treat a_t as a high-level decision that operationalizes the agentic loop by specifying the step-level information need, selecting sources, tools, and modalities to retrieve evidence, verifying intermediate results when necessary, and deciding whether to continue under explicit budgets. The interaction terminates at step T when the agent chooses to stop or reaches a hard budget. The generator g produces an output $y = g(s_T)$, where y may be text-only, or multimodal with modality-appropriate anchors. We consider both single-agent and multi-agent instantiations, where a single agent executes the task end-to-end or multiple agents coordinate by dividing labor and aggregating intermediate results (see more in Sec. 3).

3 Agentic MM-RAG Core Functions

In this section, we systematically examine the functional roles of agents in agentic MM-RAG, organizing the discussion around four core functions. As shown in Fig. 2, agentic MM-RAG operates these functions as an iterative control loop. We next review how each core function is realized across representative approaches and current open issues. A summary is presented in Table 1. Note that not all approaches implement the full set of core functions; instead, existing methods realize different subsets of agent capabilities depending on their

Agentic MM-RAG Works	Agent	Core Functions	Mod.	Learning	Code
ALFAR [3]	Single	VC	Q, C, E	ICL	[Link]
AVIS [24]	Single	F, RT, VC, SB	Q, C, E	ICL	–
CogPlanner [75]	Single	F, RT, SB	Q, C, E	ICL+SFT	–
ColPali [19]	Single	RT	Q, C, E	SFT	[Link]
CoRe MM RAG [57]	Single	F, RT, VC	Q, C, E	SFT	[Link]
Cross-modal RAG [83]	Single	F, RT, SB	C, E, G	ICL+SFT	[Link]
DeepMMSearch R1 [42]	Single	F, RT, VC, SB	Q, C, E	SFT+RL	–
DO RAG [46]	Single	F, RT, VC	C, E	ICL	–
DocReRank [62]	Single	F, VC	C, E	ICL+SFT	[Link]
Embodied RAG [67]	Multiple	F, VC, SB	C, E	ICL+SFT	–
GridMind [10]	Multiple	F, RT, SB	C, E, G	ICL	–
HM RAG [36]	Multiple	F, RT, VC, SB	C, E	ICL+SFT	[Link]
IRR-with-CBM RAG [2]	Multiple	RT, VC	Q	ICL+SFT	[Link]
KIRA [44]	Single	F, RT, VC	Q, C, E	ICL	–
MA-RAG [43]	Multiple	F	E	ICL	–
MDocAgent [22]	Multiple	F, RT, SB	C, E	ICL	[Link]
Medical-VQA RAG [56]	Multiple	F, RT, SB	Q	ICL+SFT	[Link]
MI RAG [12]	Single	F, RT, VC, SB	Q, C, E	ICL	–
Mini-o3 [30]	Single	F, RT, SB	Q, C, E	SFT+RL	[Link]
MM-RAG pipelines [18]	Single	F, RT	Q, C, E	SFT+RL	[Link]
MMSearch R1 [64]	Single	F, RT, SB	Q, C, E	SFT+RL	[Link]
mR^{2}AG [79]	Single	F, RT, VC, SB	Q, C, E	SFT	–
OminiSearch [34]	Single	F, RT, VC, SB	Q, C, E	ICL+SFT	[Link]
R1 Router [47]	Single	F, RT	C, E	RL	[Link]
ReflectiVA [13]	Single	F, RT, VC	Q, C	SFT	[Link]
RRG RAG [73]	Multiple	RT, VC	Q	ICL	–
SenseNova-MARS [11]	Single	F, RT, SB	Q, C, E	SFT+RL	[Link]
Skywork-R1V4 [81]	Single	F, RT, VC, SB	Q, C, E	SFT	[Link]
Trust-aware RAG [51]	Multiple	F, RT, VC	Q, C	ICL+SFT	[Link]
WebRAGent [4]	Single	F, RT, SB	Q, C, E	SFT	–
WebSailor [31]	Single	F, RT, SB	Q, C, E	SFT+RL	[Link]
WebWatcher [21]	Single	F, RT, SB	Q, C, E	SFT+RL	[Link]

Table 1 Representative agentic MM-RAG works (in alphabetical order). **Core Functions**: Formulate (F), Route and Use Tools (RT), Verify and Correct (VC), and Stop and Budget (SB). **Mod.** indicates where modalities beyond text appear: Query (Q), Corpus (C), Evidence (E), and Generation (G). **Learning**: In-context Learning (ICL), Supervised Fine-tuning (SFT), and Reinforcement Learning (RL). **Code** provides official implementation if available.

design goals and assumptions.

3.1 Formulate (What to Identify)

Formulate captures the agent's role in constructing retrieval targets rather than treating the query as a fixed search string. In agentic MM-RAG, this step is inherently multimodal: the agent must decide the next sub-question together with the required evidence modality and granularity, such as a text span, a table cell, a figure caption, a screenshot region, or a localized image region. These early commitments strongly constrain later routing and ranking, and are often hard to undo once the system has pursued the wrong evidence form.

In general, *Formulate* is realized through three recurring patterns: *agenda-based planning*, *failure-driven iterative refinement*, and *interaction-incentivized target construction*. Agenda-based implementations ex-

plicitly maintain and revise a target agenda that enumerates subquestions and desired evidence forms, as in CogPlanner [75]. Failure-driven implementations refine targets round by round based on partial hits, ambiguity, or dead ends, updating what to ask for and what evidence granularity is needed, as in MI-RAG [12] and AVIS [24]. Interaction-incentivized implementations strengthen multi-step target construction under search constraints through training objectives that reward effective query evolution, as in MMSearch-R1 [64] and DeepMMSearch-R1 [42]. WebWatcher [21] further extends refinement to long-horizon web research by converting tasks into increasingly specific retrieval intents.

In multi-agent settings, *Formulate* is often externalized as a shared retrieval agenda that specifies subquestions and evidence types, enabling specialists to retrieve and report evidence with stable anchors. Hierarchical delegation in HM-RAG [36] decomposes tasks into actionable subqueries, while document-heavy settings such as MDocAgent [22] emphasize page-, region-, and layout-level targets that are best specified and updated by perception and navigation agents before synthesis.

Issues. *Formulate* is not merely query decomposition; it is multimodal target engineering that reduces modality mismatch early and makes later evidence fusion, routing, and verification tractable. However, many current systems still rely on structured prompting or lightweight templates, and the resulting retrieval agendas can be underspecified or unstable. A common failure mode is that targets omit actionable constraints or anchorable evidence units (e.g., page/region-level references), forcing routing to guess toolchains and making claim–evidence alignment difficult during verification. In multi-agent settings, externalizing an agenda further raises consistency and deduplication challenges without a canonical schema for evidence units and provenance, inflating budget and weakening coverage signals.

3.2 Route and Use Tools (to Retrieve)

This function captures how agents decide where to retrieve and how to execute retrieval over heterogeneous sources and interfaces. In multimodal settings, *retrieval is inseparable from normalization*: the agent must often decide whether to browse and screenshot, run OCR, parse document layout, execute structured extraction, or query a graph, because tools produce evidence in different formats and levels of localization. Errors arise from both routing and tool choice: misrouting retrieves from the wrong source, while mis-tooling applies an incompatible mechanism, leading to modality mismatch or spurious evidence, for example using text-only search over images instead of OCR or region grounding.

Representative systems illustrate key routing patterns and clarify how routing decisions implicitly commit the system to a normalization pipeline. R1 Router [47] treats routing as a learnable, state-dependent decision across knowledge bases for stepwise retrieval, but the usefulness of routed results ultimately depends on whether raw artifacts can be normalized into evidence units that are ready for fusion. In multimodal web environments, routing is often realized through tool chaining that couples search and navigation with extraction steps, including DOM parsing, screenshot-to-text OCR, and structured field extraction. This converts retrieved pages into usable evidence units with stable anchors rather than leaving them as raw content, as demonstrated by WebRAGent [4]. In multi-agent settings, routing is commonly implemented via specialization around sources and interfaces, which further amplifies the need for explicit normalization and schema alignment. Specialist agents may return heterogeneous outputs such as OCR text with bounding boxes, layout blocks with page coordinates, or region-grounded visual descriptions, and a coordinator consolidates them into a canonical evidence registry with unified identifiers, provenance, and deduplication. GridMind [10] highlights that without normalization and schema alignment, evidence across modalities cannot be composed reliably and downstream verification lacks anchorable references.

Issues. Taken together, *Route and Use Tools* is not merely source selection; it must choose toolchains that normalize heterogeneous multimodal artifacts into anchorable evidence units for fusion and verification. However, normalization is often left implicit, so outputs may lack stable anchors or the required granularity, yielding evidence that is relevant but hard to cite and align to claims. In multi-agent settings, incompatible schemas or identifiers across specialists further complicate deduplication and provenance tracking, inflating budget through repeated retrieval and re-processing.

3.3 Verify and Correct (Evidence Use)

This function defines the agent's role in validating evidence use during generation. The agent evaluates grounding at the appropriate modality and granularity, such as image regions, layout elements, table cells, or temporal spans, while accounting for OCR errors, formatting noise, and cross-source inconsistencies. When verification fails, the agent initiates corrective actions, including revising claims, refining retrieval targets, or rerouting retrieval to alternative sources and tools. Verification outcomes are fed back into the control loop, guiding subsequent formulation and routing decisions.

Representative approaches improve grounding via *reflection, re-retrieval*, and *critique protocols*. mR^{2}AG [79] exemplifies a retrieve–rerank–reflect loop where reflection diagnoses insufficiency or mismatch and prompts targeted re-retrieval, while ReflectiVA [13] and KIRA [44] show that reflective verification can be integrated into the reasoning trace or induced training-free via prompting and retrieval reuse. Even when relevant context is present, evidence use can degrade due to attention bias or over-reliance on parametric memory; ALFAR [3] improves utilization via attention reallocation and adaptive fusion, and CoRe MM-RAG [57] emphasizes cross-source reconciliation to stabilize grounded claims, with additional auditability constraints in high-stakes domains such as radiology [2, 73]. Multi-agent designs often make verification an explicit protocol: one agent proposes evidence and intermediate claims, while another critiques support, grounding, and citation alignment, and requests targeted follow-up retrieval. Trust-aware orchestrators [51] can be viewed as verification with governance, mediating which evidence and agent outputs are accepted and triggering rerouting when support is weak or conflicting.

Issues. Overall, these works suggest that effective evidence use in MM-RAG requires both claim-level support checking and actionable correction mechanisms that couple verification to retrieval control. However, many current systems still perform verification at a coarse level and leave multimodal grounding implicit, so noise from OCR, layout ambiguity, and cross-source inconsistencies can break claim–evidence alignment at the required granularity. In multi-agent settings, this is further amplified when agents do not share a canonical evidence schema and stable anchors for regions, cells, and spans, making reconciliation and correction brittle and often triggering redundant re-retrieval.

3.4 Stop and Budget

Stopping is first-class in agentic MM-RAG because the system can always take another step, and multimodal toolchains add hidden costs from browsing, OCR, and layout parsing. DeepMMSearch-R1 and WebWatcher [21, 42] make stop decisions explicit under costed interaction, while dynamic evaluation shows fixed-depth policies are brittle and motivates self-adaptive termination [34]. In web-scale deep research, budgets are inherent and often dominate system design. In multi-agent settings, stopping becomes coordination: agents must share an evidence registry and budget to avoid duplicated work and inconsistent states. Hierarchical or shared-state control in HM-RAG and MDocAgent [22, 36] supports reliable global termination; GridMind [10] highlights that normalization and schema alignment can silently consume budget across modalities; and Embodied RAG [67] shows that memory reduces repeated retrieval but introduces synchronization overhead that must be managed for stable stopping.

Issues. Although recent approaches make stopping and costed interaction explicit, key gaps remain. Most policies still use coarse step budgets or proxy costs and do not reliably model multimodal tool overhead from browsing, OCR, and layout parsing. Termination is often driven by heuristics or answer confidence, with limited criteria for evidence sufficiency and marginal utility, leading to premature stopping or redundant over-search.

3.5 Learning & Optimizing Agent Behavior

The agent's behavior in agentic MM-RAG is governed by a policy π_θ. How this policy is induced and optimized plays a central role in system efficiency. Existing approaches primarily rely on three paradigms: (1) *in context learning* elicits agent behavior at inference time through prompts, instructions, and exemplars without updating model parameters; (2) *supervised fine-tuning* trains the policy from logged agent traces; and (3) *reinforcement learning* optimizes long horizon loop control under explicit cost and reliability objectives.

In-context Learning (ICL). This is a lightweight approach: shaping inference-time functions by prompts, tool schemas, and exemplars. Prompts specify the action language (rewrite, search, browse, parse, verify, stop) and the evidence contract (cite and align modality-specific anchors), providing a modular scaffold that is easy to adapt to new tools and interfaces. Planning-oriented prompting can stabilize *Formulate* and *Stop* behaviors by drafting a retrieval plan and iteratively revising it based on partial observations, as in CogPlanner [75]. Iterative MRAG patterns, such as MI-RAG [12], can also be expressed in a purely prompted manner by maintaining an explicit evidence gap and generating increasingly targeted multi-queries. In open-world visual information seeking, AVIS [24] illustrates that structured protocols and memory-based prompting improve tool-use coherence even without parameter updates. Overall, ICL-style methods are low-cost and flexible, but their robustness is bounded by the base model's intrinsic tool competence and grounding reliability.

Supervised Fine-tuning (SFT). SFT is the most common approach for making agentic behaviors stable and reproducible. The core idea is to fine-tune models on interaction trajectories, which are state conditioned action sequences that include query rewrites, tool calls, retrieved evidence, critique and verification actions, and termination decisions. Deep research and multimodal web search systems often begin with a cold start SFT stage that teaches multi-turn search, browsing, and extraction patterns before any additional optimization. As in DeepMMSearch-R1 [42] and WebWatcher [21]; recent work also shows that sufficiently curated trajectories can yield strong agentic behaviors with SFT alone (e.g., Skywork R1V4 [81]). SFT is not limited to training the controller: it also trains retrieval-side modules and verifiers that directly support *Route/Tool* and *Verify/Correct*. For example, DocReRank [62] improves multimodal reranking through hard negative query generation; ReflectiVA [13] and mR^{2}AG [79] inject supervision for retrieval gating and reflection; CoRe-MM-RAG [57] emphasizes cross-source reconciliation signals; and large-scale tool-use SFT such as ToolLLM [50] improves tool invocation formats and chaining competence. Despite its effectiveness, SFT has two recurring limitations. First, high-quality supervision is costly to obtain and fragile to collect: trajectories often require strict logging or re-executing tools so that the recorded actions faithfully match the observed outputs, which can be expensive and hard to scale across heterogeneous interfaces. Second, SFT tends to inherit the behavioral support of its training traces and may generalize poorly when the tool API, web layout, domain distribution, or multimodal evidence patterns shift. Together, these limitations motivate objective-driven optimization beyond trace imitation, where the agent is trained to maximize grounded utility under explicit interaction budgets. Reinforcement learning is a common instantiation of this idea.

Reinforcement learning (RL). This objective-driven methods treat the iterative process as a sequential decision process and optimize rewards that combine answer quality with efficiency penalties (tool calls, page visits, latency), directly targeting *Stop/Budget* and discouraging redundant retrieval. Search-incentivized training exemplifies this paradigm: MMSearch-R1 [64] and DeepMMSearch-R1 [42] reward effective multi-step search under interaction constraints, while multimodal agentic search systems such as SenseNova-MARS [11] and Mini-o3 [30] emphasize longer-horizon interleaved perception–search behaviors. Beyond single-agents, multi-agent RL formulations jointly optimize specialized roles, aligning *Formulate/Route/Verify*decisions with global grounded performance [8]. However, RL for agentic MM-RAG is often harder to deploy than it appears: designing reliable rewards for grounding and tool correctness is non-trivial, long-horizon credit assignment can be unstable, and exploration may incur substantial tool costs or unsafe failure modes in open-world environments. As a result, many practical systems use SFT for a strong and safe initialization, and apply limited RL-style refinement primarily to improve budgeted stopping and reduce redundant retrieval.

4 Evaluation

To diagnose agentic MM-RAG behavior beyond final-answer correctness, we adopt an evaluation perspective aligned with the core functions. Specifically, the evaluations framework includes: (1) *Retrieval Specification* measures whether the system specifies step-level information needs prior to retrieval and makes the required evidence modality and granularity explicit, aligned with *Formulate* function; (2) *Retrieval Quality* measures whether the system selects appropriate sources and mechanisms to acquire evidence and localizes it to the correct citable units, aligned with *Route & Use Tools* function; (3) *Evidence Grounding in Generation* measures whether the final response is grounded in retrieved evidence with correct citation or anchor alignment, aligned with *Verify & Correct*; and (4) *Stopping Optimality* measures whether the system terminates under budget

Eval Dimension	Benchmark	Venue	Metrics	Key Contributions
Retrieval Specification	CogPlanner [75]	SIGIR-AP	Claim-level (Precision, Recall), token-level (F1)	Tailored to the MRAG planning task.
	MC-Search [45]	NeurIPS WS	Acc($F1, \Delta F1$, Golden$F1$), Chain Alignment(LJ, HPS, RD)	Structured multi-hop reasoning chains.
	MM-BrowseComp [32]	arXiv	Accuracy, Average Checklist Score (AVG CS)	Multimodal questions with image and text.
Retrieval Quality	LongDocURL [16]	ACL Main	Generalized Accuracy Score	Long-document understanding and reasoning.
	M^2RAG [39]	ACM MM	BERTScore, CIDEr, ROUGE, Accuracy, F1, FID	Integrate image & text during retrieval and generation.
	MMLongBench [61]	NeurIPS	Substring Exact Match (SubEM), Accuracy, LLM-score	Long-context vision-language task.
	UDA [26]	NeurIPS D&B	F1-score, LLM-score, Exact-Match, LCS	Evaluation of various RAG components.
	VDocRAG [54]	CVPR	nDCG@k, ANLS, Relaxed Accuracy F1	Unified open-domain dataset with diverse documents.
Evidence Grounding	CaptionQA [70]	arXiv	Accuracy, Cannot ratio	Utility-oriented caption evaluation.
	FinRAGBench-V [82]	EMNLP	nDCG@k, Recall@k, MRR@k, ROUGE, Acc, Precision	Multimodal financial data with visual citations.
	MRAMG-Bench [74]	SIGIR	Context/Visual Rec, LLM-scores, Image P/R/F1, ROUGE-L	Hierarchical difficulty levels, Order-based reasoning.
	RAG-IGBench [78]	NeurIPS D&B	ROUGE-1, Edit Distance, Kendall/CLIP/Alignment Score	Interleaved image-text generation.
Stopping Optimality	BrowseComp-Plus [9]	arXiv	Acc, Recall, Search Calls, Calibration Error, nDCG@k	W/ supporting docs and mined challenging negatives.
	CRAG-MM [60]	arXiv	Truthfulness, Accuracy, Missing Rate, Hallucination Rate	Multi-modal multi-turn conversations.
	Dyn-VQA [34]	arXiv	F1-Recall, LLM/Human-based Accuracy	Retrieval strategies variable in query, tool and time.

Table 2 Representative benchmarks organized by four evaluation dimensions.

constraints by trading off marginal evidence gains against interaction cost, aligned with *Stop & Budget*. We summarize benchmark coverage across these evaluation dimensions in Table 2.

Retrieval Specification. This evaluation includes sub-question coverage, the intended evidence modality and granularity (passage/figure/table-cell/region/timestamp), and routing intent, namely *from where* to retrieve and *with what* mechanism. Process-supervised resources are particularly valuable here: MC-Search [45] provides hop-wise annotated reasoning chains (sub-questions, retrieval modalities, intermediate answers) and chain-level metrics to evaluate planning fidelity beyond answer accuracy; CogPlanner [75] formalizes an MRAG planning task and introduces CogBench with recorded iterative plans to score decision quality; MM-BrowseComp [32] further stresses multimodal browsing by pairing each question with a verified checklist, enabling fine-grained audits of whether the agent identified the necessary multimodal evidence and reasoning steps. Quantitatively, *Retrieval Specification* can be measured via hop/step alignment or checklist coverage when available, and via efficiency proxies such as detours or redundant queries before reaching supporting evidence when gold intermediate decisions are absent.

Retrieval Quality. Retrieval Quality evaluates whether the system actually acquires the required evidence from the intended sources and localizes it at the appropriate granularity in multimodal substrates, such as a page, region, cell, or timestamp. Document-centric suites are representative: UDA [26] evaluates RAG over lengthy, unstructured real-world HTML/PDF documents with expert QA, highlighting retrieval and parsing as key bottlenecks; LongDocURL [16] explicitly integrates long-document understanding, numerical reasoning, and cross-element locating to stress "find-then-reason-then-locate" behavior; MMLongBench [61]

provides controlled long-context settings (standardized input lengths) that make retrieval and context packing decisions comparable; and VDocRAG [54] targets visually rich documents by treating documents as images to reduce information loss from text-only parsing, while also introducing OpenDocVQA as an open-domain document VQA resource. We further include M2RAG [38] as a multi-modal RAG benchmark that evaluates how well MLLMs leverage retrieved multi-modal contexts across open-domain tasks such as captioning, QA, fact verification, and image reranking, complementing document-centric suites with a task-diverse view of evidence acquisition and localization. Typical metrics include standard IR measures (Recall@K, nDCG, and MRR) for evidence recall, as well as locating accuracy when the target is a specific page, region, or cell.

Evidence Grounding in Generation. This evaluates whether the final response is faithfully supported by the retrieved evidence and whether citations or anchors align to modality-appropriate units, such as text spans, regions, cells, or timestamps, rather than being merely plausible. Benchmarks that make attribution explicit are especially informative. CaptionQA [70] provides a utility-based grounding test by measuring whether captions retain enough visual evidence to answer multiple-choice questions. FinRAGBench-V [82] requires *visual citation* in financial QA and proposes automatic citation evaluation to score traceability; RAG-IGBench [78] evaluates retrieval-grounded *interleaved* image-text generation with metrics that capture both unimodal quality and cross-modal consistency; and MRAMG-Bench [74] moves beyond text-only answers by defining MRAMG (retrieve multimodal evidence and generate multimodal outputs), coupled with statistical and LLM-based metrics for multimodal generations. Across these settings, evaluation should penalize unsupported claims and misaligned citations, and reward *Verify/Correct* behaviors that detect mismatches or contradictions and trigger re-retrieval.

Stopping Optimality. Dynamic and interactive benchmarks make this evaluation axis concrete. Dyn-VQA [34] introduces "dynamic" questions (e.g., rapidly changing answers, multimodal knowledge needs, multi-hop) that expose rigid fixed-step retrieval policies and motivate self-adaptive planning; MMSearch-Plus [55] enforces long-horizon multimodal browsing by requiring agents to extract weak localized visual cues and iteratively cross-validate under noisy retrieval, explicitly stressing provenance-aware tool use; BrowseComp-Plus [9] improves reproducibility for deep-research evaluation by using a fixed curated corpus with human-verified supporting documents and hard negatives, enabling controlled accuracy–cost comparisons; and CRAG-MM [60] targets multimodal, multi-turn user scenarios (including egocentric images) with paired corpora/APIs for single-source, multi-source, and multi-turn retrieval, making statefulness and budget management unavoidable. Metrics should jointly report cost signals (e.g., tool calls, steps, latency, and tokens) together with performance under fixed budgets, ideally presenting an accuracy–cost trade-off rather than a single scalar.

Guidance and Gaps. We recommend using Table 2 as a diagnostic suite rather than evaluating with a single score: first validate *Retrieval Specification* with process-supervised or checklist-based benchmarks, then stress *Retrieval Quality* on document- and layout-heavy suites with localization (page/region/cell/timestamp), next test *Evidence Grounding* with citation/anchor-sensitive benchmarks, and finally report *Stopping Optimality* as an accuracy–cost trade-off under explicit budgets (steps/tool calls/tokens/latency). However, current benchmarks must be substantially advanced to fully support agentic MM-RAG. First, domain-specific benchmarks are needed to reflect realistic knowledge substrates and toolchains (e.g., scientific documents, charts, scanned forms, or video timelines). Second, today's benchmarks remain fragmented across tasks and environments, making it difficult to evaluate end-to-end agent behavior within a unified control loop. Moreover, practical factors such as noisy OCR and layout pipelines, cross-source inconsistencies, tool execution failures, and budget-induced truncation are still under-measured. Addressing these gaps is essential for developing benchmarks that faithfully capture the challenges faced by agentic MM-RAG systems in real-world settings.

5 Discussions and Future Opportunities

Applications. Agentic MM-RAG is increasingly adopted in real-world settings that require iterative evidence seeking, multimodal grounding, and budgeted decision making. Representative applications span scientific research, medicine, education, law, and sports and wellness. In scientific workflows, agentic systems decompose research goals into retrieval-oriented subtasks and iteratively refine queries to support dataset construction and grounded drafting, e.g., CollEX [52], Quest2DataAgent [71], and AI Scientist [40]. In medicine, where

reliability is critical, agentic MM-RAG integrates multimodal clinical evidence and curated knowledge to reduce hallucinations in diagnosis and QA, including MMed-RAG [65] and Patho-AgenticRAG [80], as well as multi-agent consistency control for report generation [2, 73]. In education, agentic retrieval over diverse instructional materials supports robust tutoring and summarization under noisy queries, such as Uni-Retrieval [29], Parrot [14], and KA-RAG [20]. In legal affairs, agentic pipelines emphasize clause- and evidence-level retrieval with citation-faithful long-form generation for auditable assistance, e.g., jurisdiction-specific systems [23] and PAKTON [49]. Finally, in sports and wellness, agentic MM-RAG enables personalized, long-horizon assistance grounded in multimodal signals and structured databases, including video-based coaching and programmatic retrieval for sports analytics [59, 63], and evidence-grounded health recommendation via expert-curated knowledge [68, 69].

Challenges and Future Directions. Agentic MM-RAG is advancing rapidly from proof-of-concept systems toward real world deployment. Yet, scaling these systems remains difficult because failures in one stage of the loop can propagate and compound across iterations.

Reliability & Safety. Reliability in agentic MM-RAG is not only about reducing intrinsic hallucinations [6], but also about procedural correctness across intent inference, modality selection, tool invocation, and faithful evidence use. Multimodal settings introduce modality-dependent failures, including OCR and layout noise, region- or cell-level misgrounding, and cross-modal or cross-source conflicts [48, 58]. These errors are often subtle, because the "correct" support may reside in a specific page region, table cell, or time span, and mistakes early in the loop can bias later retrieval and verification. Open-world retrieval further raises safety risks such as prompt or tool injection and retrieval poisoning [28, 76, 77], which can directly corrupt the evidence registry and downstream reasoning. Future systems should treat grounding, conflict handling, and safety constraints as a unified reliability layer, combining provenance-aware evidence selection, calibrated uncertainty, and targeted correction that localizes unreliable steps instead of global regeneration. Multi-agent designs are also promising here: role specialization and cross-checking can reduce confirmation bias and catch procedural errors earlier, provided that evidence acceptance and overrides are governed by explicit protocols.

Evaluation & Benchmarks. Current evaluations often under-measure agentic and multimodal failure modes, such as choosing the right modality and granularity, resolving modality conflicts, and stopping for the right reasons under tool costs. Moreover, answer-level accuracy can mask brittle behaviors, for example when correct outputs arise from lucky retrieval rather than robust routing and verification. Future benchmarks should emphasize trajectory-level diagnosis that attributes errors to specific decisions, tools, and evidence units, and should make multimodal stressors first-class, including misalignment across space and time, noisy extraction, and cross-source inconsistency. It is also important to evaluate controllable behaviors, such as whether systems can abstain, request clarification, or trigger targeted re-retrieval when support is weak, rather than always committing to a single final response.

Cost & Latency. Iterative retrieval and multi-agent coordination can improve coverage but increase token use, tool overhead, and end-to-end latency, especially in multimodal pipelines where preprocessing often dominates cost. Compared with text-only RAG, substantial time is spent on browsing and rendering pages, OCR and layout parsing, and region-level grounding, so deeper loops can become disproportionately expensive. Existing adaptive retrieval and trajectory compression [27, 66] only partially address redundant tool calls and repeated multimodal extraction in agentic settings. Promising directions include budget-aware step selection and early stopping, caching and deduplication of intermediate multimodal artifacts across iterations and agents, and learned or calibrated cost models that trade off marginal evidence gain against tool overhead.

6 Conclusion

Agentic MM-RAG is becoming a key approach for grounding and controlling vision–language models via iterative, policy-driven decisions over retrieval, verification, and budgets. Our framework presents an agent-centric view that formalizes the setting, organizes prior work around four core functions, and reviews evaluation practices that diagnose failures beyond final answer quality. We expect this framework to serve as a foundation for more principled system design, learning, and benchmarking, and to accelerate the development of robust, efficient, and deployable agentic MM-RAG systems.

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